<u>Lecture - 19</u>

11 March 2026

Recap: LTI Systems

Defn: A system is called linear time invariant or linear translation invariant or linear shift invariant if the system is both linear and time invariant.

To check for LTI property:

1. Check for linearity
2. Check for time invariance

System must satisfy both.

If T is an LTI system

Suppose $x_1 \longrightarrow \boxed{T} \longrightarrow y_1$ and $x_2 \longrightarrow \boxed{T} \longrightarrow y_2$

$$\alpha_1 x_1(t-t_1) + \alpha_2 x_2(t-t_2) \longrightarrow \boxed{T} \longrightarrow \alpha_1 y_1(t-t_1) + \alpha_2 y_2(t-t_2)$$

Examples of LTI systems

1. $y = x * h$ for some known signal h (Homework)

2. $y(t) = \sum_{i=1}^{K} a_i \, x(t-t_i)$ (Homeworl

I LTI Systems are Characterized by Convolution

If T is an LTI system (under some mathematical conditions[†]):

there exists a unique signal h associated with T

such that

$$T(x) = x * h$$

$$x \longrightarrow \boxed{T} \longrightarrow x * h$$

- h is called the impulse response of T
- h is a property of the system T (not of the input x)

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† Ref: Stein & Weiss "Introduction to Fourier Analysis on Euclidean Spaces."

Chapter 1, Theorem 3.16.

New notation

$$x \longrightarrow \boxed{T} \longrightarrow y$$

$$x \longrightarrow \boxed{h} \longrightarrow y = x * h$$

- Knowledge of h completely describes the system T

Consequences: (Sec 2.3 of OWN)

(1) Cascade or Series Interconnection of LTI systems

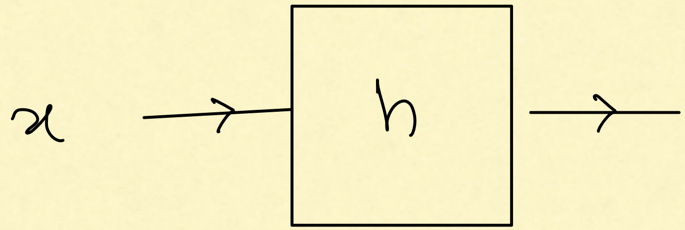
$$x \longrightarrow \boxed{T_1} \xrightarrow{\;w\;} \boxed{T_2} \longrightarrow \quad y = T_2(w) = T_2(T_1(x))$$

$$x \longrightarrow \boxed{h_1} \xrightarrow{\;w\;} \boxed{h_2} \longrightarrow y$$

$$w = x * h_1$$

$$y = w * h_2 = (x * h_1) * h_2$$
$$= x * (h_1 * h_2)$$

$\therefore$ The overall system is:

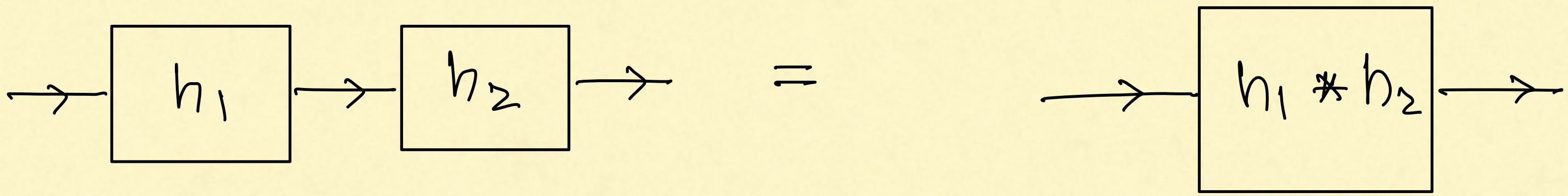


$y = x * h$

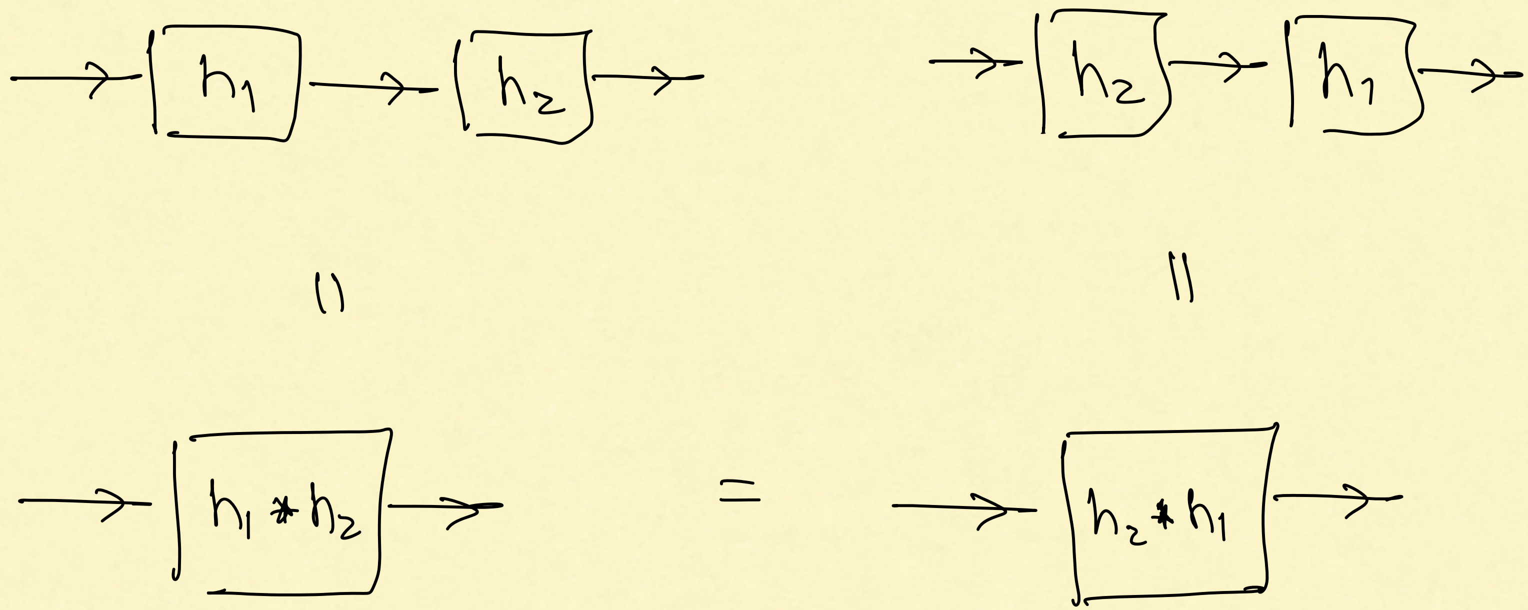
where $h = h_1 * h_2$

Since the input-output relation is given by convolution,

the overall system is LTI.



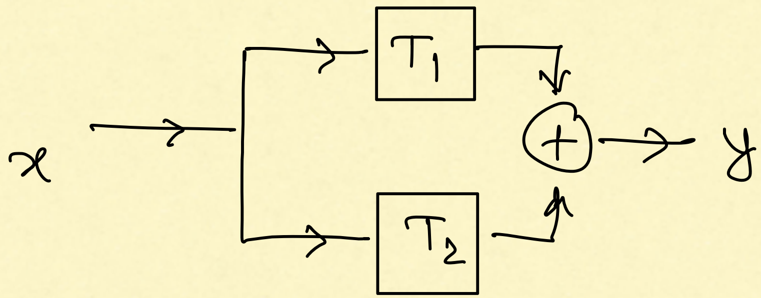
since $h_1 * h_2 = h_2 * h_1$, we have



The order in which we connect the LTI systems in cascade does not affect the final output.

(But it will affect the intermediate outputs in the cascade).

2 Parallel Interconnection



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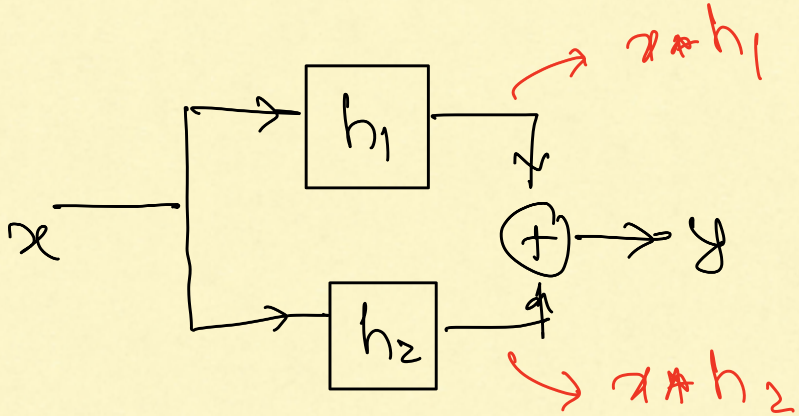
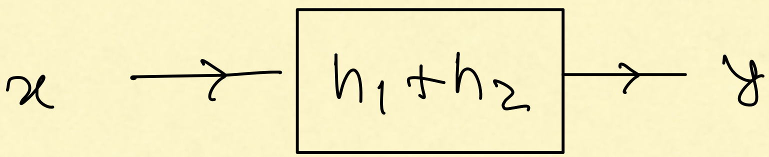
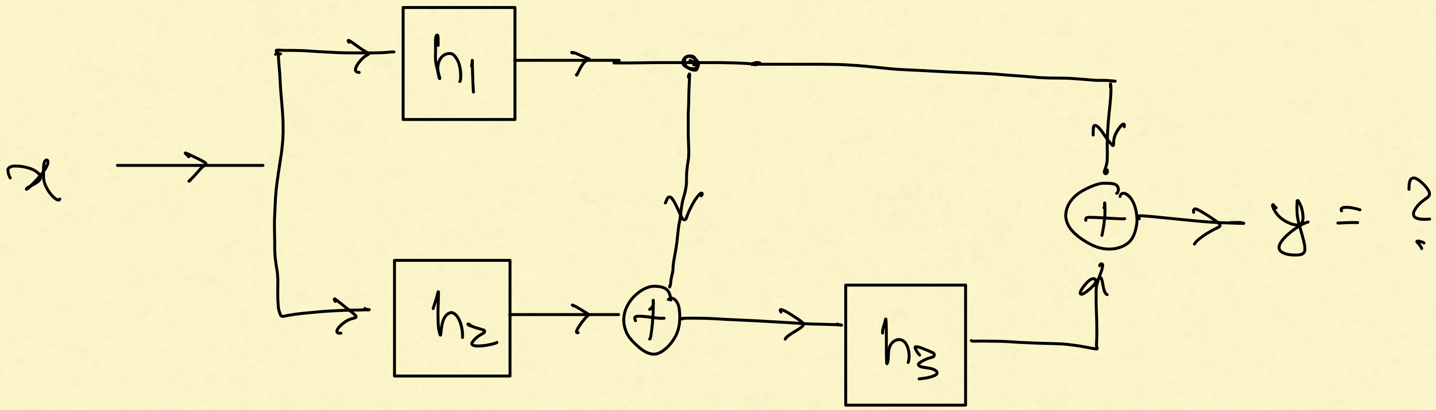


$\therefore\ y = x * h_1 + x * h_2 = x * (h_1 + h_2)$

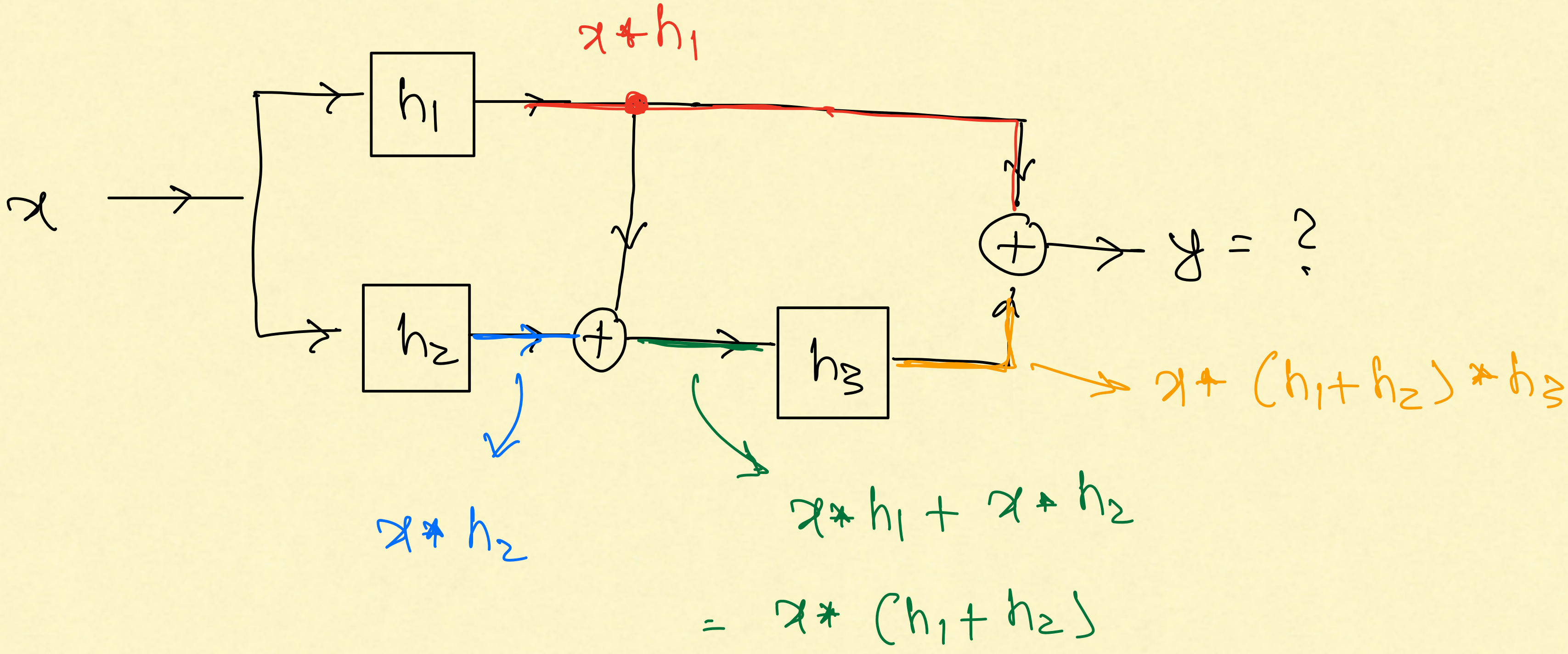
$\therefore$ This system is equivalent to:



<u>Example</u>: Any combination of cascade and parallel interconnections of LTI systems is an LTI system.



Example :

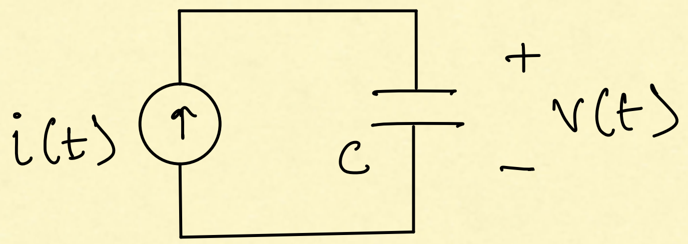


$$\therefore\ y = x * (h_1 + h_2) * h_3 + x * h_1$$

$$= x * [h_1 * h_3 + h_2 * h_3 + h_1]$$

The impulse response of the overall system: $h_1 * h_3 + h_2 * h_3 + h_1$

Example 1



$$v(t) = \frac{1}{C} \int_{\tau=-\infty}^{t} i(\tau)\,d\tau \qquad \text{—— } (1)$$

Consider : $i \rightarrow \boxed{T} \rightarrow v$

What is the impulse response ?

Hint : Write equation (1) as a convolution

$$v(t) = \int_{\tau=-\infty}^{+\infty} i(\tau)\, h(t-\tau)\,d\tau .$$

$$v(t) = \int_{\tau=-\infty}^{t} \frac{1}{C} i(\tau)\, d\tau$$

$$= \int_{\tau=-\infty}^{+\infty} \frac{1}{C} i(\tau) \cdot \mathbb{1}\{\tau \le t\}\, d\tau$$

indicator function

$$\mathbb{1}\{\tau \le t\} = \begin{cases} 1 & \text{if } \tau \le t \\ 0 & \text{if } \tau > t \end{cases}$$

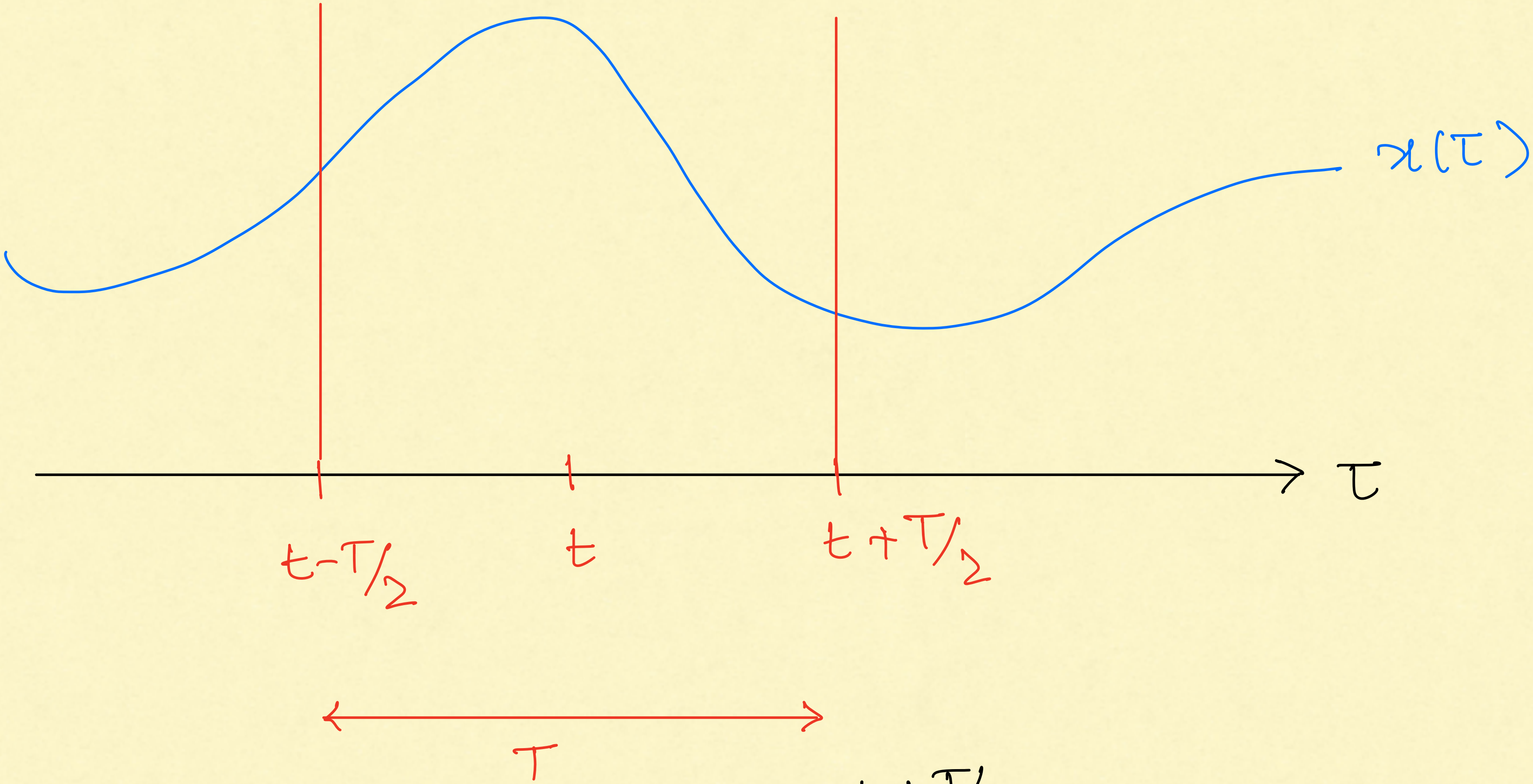
$$= \int_{\tau=-\infty}^{+\infty} \frac{1}{C} i(\tau)\, u(t-\tau)\, d\tau$$

$$= \int_{\tau=-\infty}^{+\infty} i(\tau) \cdot \frac{1}{C} u(t-\tau)\, d\tau$$

$$= i * \frac{1}{C} u.$$

$$\Longrightarrow \quad h(t) = \frac{1}{C} u(t). = \begin{cases} \frac{1}{C} & \text{if } t \ge 0 \\ 0 & \text{if } t < 0 \end{cases}$$

Example 2 : Moving average filter



$$y(t) = \frac{1}{T} \int_{\tau = t - T/2}^{t + T/2} x(\tau)\, d\tau$$

$$x \to \boxed{T} \to y$$

$$y(t) = \frac{1}{T} \int_{\tau = t - T/2}^{t + T/2} x(\tau)\, d\tau$$

$$= \frac{1}{T} \int_{\tau = -\infty}^{+\infty} x(\tau) \cdot \mathbb{1}\left\{ \tau \in [t - T/2, t + T/2] \right\} d\tau$$

$$= \frac{1}{T} \int_{\tau = -\infty}^{+\infty} x(\tau)\, \mathbb{1}\left\{ |t - \tau| \leq T/2 \right\} d\tau$$

Define $h(t) = \mathbb{1}\left\{ |t| \leq T/2 \right\} \cdot 1/T$

$$\Longrightarrow \quad h(t - \tau) = \mathbb{1}\left\{ |t - \tau| \leq T/2 \right\} \cdot 1/T$$

$$\therefore \quad y(t) = \int_{-\infty}^{+\infty} x(\tau)\, h(t - \tau)\, d\tau = x * h$$

Questions

1. How do we find the impulse response of a given LTI system?

2. Why are LTI systems characterized by convolution?

II Impulse Response of Identity System (Simplest system)

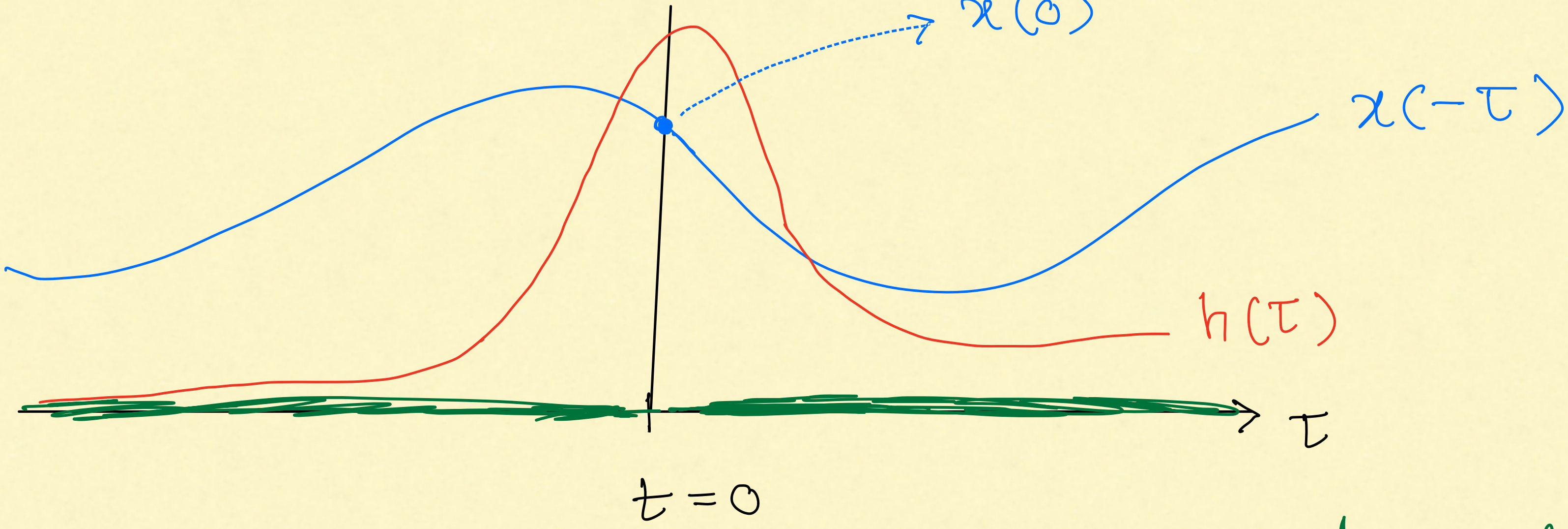
$$x \longrightarrow \boxed{T} \longrightarrow y(t) = x(t)$$

$$x \longrightarrow \boxed{h} \longrightarrow x * h = x$$

What is h?

$$y(t) = \int_{-\infty}^{+\infty} h(\tau)\, x(t-\tau)\, d\tau$$

$$\Rightarrow \quad y(0) = \int_{-\infty}^{+\infty} h(\tau)\, x(-\tau)\, d\tau$$



We want h to be such that: (This is an informal argument)

$$\int_{-\infty}^{+\infty} h(\tau)\,x(-\tau)\,d\tau = x(0) \quad \text{for any input } x$$

$\Rightarrow$ No weightage for $\tau \neq 0 \Rightarrow h(\tau) = 0$ for $\tau \neq 0$

$$\Rightarrow \int_{-\infty}^{+\infty} h(\tau)\,x(-\tau)\,d\tau = \int_{0^-}^{0^+} h(\tau)\,x(-\tau)\,d\tau = 0$$

$\Rightarrow$ No such h exists.

Since no function can come to our rescue we resort to:

Generalized functions or **Distributions**

What is a distribution?

A distribution is an operator that maps a signal to a complex number.

$$\langle \text{distribution}, \text{signal} \rangle = \text{complex number}$$

assuming some mathematical restrictions on signal.

(This is the notation used in Osgood's textbook)

Note:

For a precise mathematical definition,

we require a distribution to act on

signals linearly and continuously. That is:

1. $\langle \text{dist.}, \alpha_1 \text{signal}_1 + \alpha_2 \text{signal}_2 \rangle$

$= \alpha_1 \langle \text{dist.}, \text{signal}_1 \rangle + \alpha_2 \langle \text{dist.}, \text{signal}_2 \rangle$

2. If we change the signal a little, the value of $\langle \text{distribution}, \text{signal} \rangle$ changes by a small amount only.

References (advanced) for distributions:

1. Chapter 4 of Osgood's textbook.
2. Chapter 5 of Bracewell, "The Fourier Transform and its Applications".
3. M. J. Lighthill, "Fourier Analysis and Generalized Functions".

Example 1 — Dirac's Delta Distribution δ

(incorrectly called Dirac's Delta "Function")

Definition: $\langle \delta, x \rangle = x(0)$

distribution

signal

scalar result

Example: If $x(\tau) = 5\cos(2\pi t) + 3$

$$\langle \delta, x \rangle = x(0) = 5 + 3 = 8.$$

Example 2a $\quad$ δ based at t_0 : δ_{t_0}

$$\langle \delta_{t_0}, x \rangle = x(t_0)$$

- If $t_0 = 0$: δ_{t_0} and δ are same
- If $t_0 \neq 0$: δ_{t_0}, δ are distinct distributions
- In a sense, δ_{t_0} is a "time-shift" of δ.

Example 2b : Scaled Delta: $a\delta$

If 'a' is a scalar:

$$\langle a\delta, x\rangle = a\langle \delta, x\rangle$$

$$= a\,x(0).$$

Similarly, $\langle a\delta_{t_0}, x\rangle = a \cdot x(t_0).$

Example 3 : Linear Combination of shifted delta's

Define a distribution T as:

$$T = a_1 \delta_{t_1} + a_2 \delta_{t_2} + \cdots + a_k \delta_{t_k}$$

$a_1, \ldots, a_k, t_1, \ldots, t_k$ are constants.

$$\langle T, x \rangle = \langle a_1 \delta_{t_1} + \cdots + a_k \delta_{t_k}, x \rangle$$

$$= \langle a_1 \delta_{t_1}, x \rangle + \cdots + \langle a_k \delta_{t_k}, x \rangle$$

$$= a_1 \langle \delta_{t_1}, x \rangle + \cdots + a_k \langle \delta_{t_k}, x \rangle$$

$$= a_1 x(t_1) + \cdots + a_k x(t_k)$$

Example 4 : Shah Distribution or Dirac Comb

$$Ш = \sum_{k=-\infty}^{+\infty} \delta_k = \dots + \delta_{-1} + \delta_0 + \delta_1 + \delta_2 + \dots$$

$$\langle Ш, x \rangle = \sum_{-\infty}^{+\infty} \langle \delta_k, x \rangle$$

$$= \sum_{k=-\infty}^{+\infty} x(k)$$

$$= \dots + x(-1) + x(0) + x(1) + x(2) + \dots$$